

Mary Sirisha Kasipathi

Data Engineer (AI & ML)

SUMMARY

Data Engineer with **10+ years of experience** designing, building, and operating **scalable data platforms, ETL pipelines, and ML data infrastructure** across fintech, retail, healthcare, and telecom domains. Currently focused on **AI/ML data systems at Visa**, supporting **fraud detection and risk scoring models processing billions of global payment transactions**. Strong expertise in **cloud-native architectures (AWS, Azure)**, **big data frameworks (Apache Spark, PySpark, Kafka)**, and **MLOps workflows (SageMaker, MLflow, feature engineering)**. Proven track record in building **batch and real-time streaming pipelines**, optimizing **data lake architectures**, and implementing **data quality and validation frameworks** to ensure reliable analytics and production ML operations. Experienced in collaborating with **data scientists, ML engineers, and product teams** in **Agile environments** to translate complex business requirements into robust, scalable data solutions.

SKILLS

Programming & Scripting: Python (PySpark, pandas), Scala, SQL, Shell Scripting, REST APIs

Cloud Platforms: AWS (S3, EMR, SageMaker, Glue, Lambda, EC2, IAM, VPC, Athena, CloudWatch), Azure (Databricks, Synapse Analytics, Data Factory, Event Hubs, ADLS Gen2, Azure Monitor, Azure DevOps)

Databricks&Lakehouse: Databricks, Delta Lake, Unity Catalog, Databricks Notebooks, Databricks Structured Streaming, Databricks MLflow

Big Data & Streaming Technologies: Apache Spark, PySpark, Spark Structured Streaming, Apache Kafka, Apache Airflow, Hadoop (HDFS, Hive, MapReduce)

Databases & Storage: PostgreSQL, Oracle, SQL Server, MySQL, MongoDB, Teradata, Delta Lake, Parquet, Amazon S3, ADLS Gen2

Machine Learning & MLOps: AWS SageMaker, MLflow, Databricks MLflow, Feature Engineering, Model Training Pipelines, Experiment Tracking, Dataset Versioning, Model Registry, Great Expectations, Data Validation Frameworks

DevOps & CI/CD : Git, GitHub, Jenkins, Terraform, Ansible, Azure DevOps, CI/CD Pipelines

Project Management & Methodologies: Agile, Scrum, SDLC, Jira

Monitoring & Observability : AWS CloudWatch, Azure Monitor, Datadog, Great Expectations, Data Validation Frameworks

Methodologies: Agile/Scrum, SDLC, ETL/ELT Development, Data Modeling, Dimensional Modeling, Technical Documentation, Jira, Confluence

EXPERIENCE

Visa | Data Engineer (AI & ML)

WA | January 2025 – Present

- Design and operate scalable ML data pipelines on **Databricks** and **AWS** to support fraud detection and risk scoring systems processing billions of global payment transactions.
- Build **Spark-based batch pipelines** using Databricks, **Delta Lake**, and **AWS EMR** to preprocess transaction data, merchant attributes, and behavioral signals for model training and inference, ensuring **ACID compliance** and data versioning.
- Implement **feature engineering frameworks** in Python and PySpark on Databricks, generating time-windowed aggregates, velocity features, and encoded variables for production fraud models, with **Unity Catalog** for governance and lineage tracking.
- Develop **real-time streaming pipelines** using **Kafka** and **Databricks Structured Streaming** to enrich authorization events with merchant risk indicators, geo-signals, and historical context for low-latency model scoring.
- Integrate **MLOps data workflows** using **Databricks MLflow**, **AWS SageMaker**, and Delta Lake to enable dataset versioning, experiment tracking, model registry, and promotion of production ML artifacts.
- Enforce **data quality, schema validation, and drift monitoring** using **Great Expectations** and Databricks monitoring tools to ensure reliability of ML features in production.
- Collaborate with data scientists and ML engineers via Databricks notebooks to refine feature requirements, resolve data issues, and improve performance and stability of ML services.

Walmart | Data Engineer

AR | September 2022 – December 2024

- Built and maintained **Azure data pipelines** ingesting sales, inventory, and supply chain data from **4,500+ retail stores** using **Azure Data Factory** and **Databricks (PySpark)** to deliver analytics-ready datasets.
- Implemented **near real-time streaming pipelines** using **Kafka** and **Azure Event Hubs**, improving inventory visibility and enabling faster stock replenishment decisions across the retail network.
- Developed **dimensional data models** in **Azure Synapse Analytics** and optimized **Delta Lake** workloads using partitioning and caching, reducing query execution time by **~60%**.
- Automated **third-party vendor data ingestion** using **Python** and **REST APIs** into **ADLS Gen2**, and implemented **data quality checks and monitoring** with **Python** and **Azure Monitor** to ensure SLA compliance.

Accenture | Senior Software Engineer

India | January 2016 – August 2019

- Led migration of **legacy ETL platforms** to **AWS-based cloud architectures** for enterprise healthcare clients (Cigna, Express Scripts), modernizing data platforms supporting **millions of member records**.
- Designed and built **scalable batch and streaming data pipelines** using **Apache Spark, PySpark, EMR, Kafka, and Spark Structured Streaming**, reducing claims adjudication processing time by **~50%**.
- Implemented secure and scalable **AWS infrastructure** including **EC2, S3 (lifecycle policies), IAM, VPC**, and S3-based **data lake architectures** optimized for Athena ad-hoc analytics.
- Developed **reusable, parameterized PySpark frameworks** with dynamic schema handling, enabling consistent data processing across multiple business domains.
- Established **CI/CD pipelines using Jenkins and Git**, automating build, test, and deployment of Spark applications across **development, QA, and production** environments.
- Integrated **data quality and validation frameworks** within **Spark jobs** to enforce data completeness, consistency, and business rule compliance, while **leading technical design discussions**, mentoring **4–6 engineers**, conducting code reviews, and collaborating with product owners in **Agile/Scrum** environments to deliver reliable data solutions.

- Designed, built, and maintained **batch data pipelines** processing **millions of healthcare and pharmaceutical records**, including claims data, patient demographics, and clinical trial datasets, supporting regulatory reporting and compliance needs.
- Developed and optimized **data extraction, transformation, and loading (ETL) workflows** using **Python, SQL, and Shell scripting** on Linux, integrating data from **Oracle, SQL Server, and flat file sources**.
- Built and enhanced enterprise **ETL pipelines** using **IBM DataStage and Informatica PowerCenter**, delivering curated datasets to centralized data warehouses for analytics and business intelligence teams.
- Processed large-scale datasets using the **Hadoop ecosystem (HDFS, Hive, MapReduce)**, supporting early big data initiatives for **telecom billing and network usage analytics**.
- Implemented **data quality checks, validation rules, and reconciliation logic** to ensure accuracy, completeness, and consistency across healthcare and customer datasets, reducing downstream reporting defects.
- Partnered with **business analysts, QA teams, and downstream data consumers** in waterfall and early Agile environments to deliver data pipelines aligned with **project requirements and SLAs**.
- Troubleshoot and resolved **production data pipeline failures** by analyzing logs, identifying root causes, and implementing fixes in **on-premise environments**, improving pipeline stability.
- Authored and maintained **technical documentation**, including data mappings, ETL designs, and operational runbooks, enabling long-term maintainability and smooth client transitions.